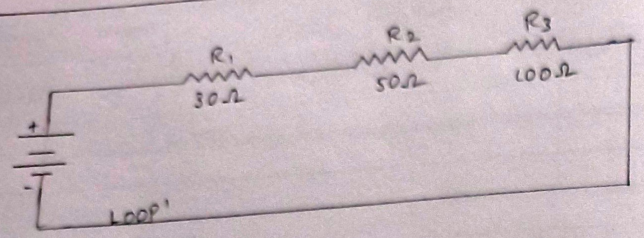
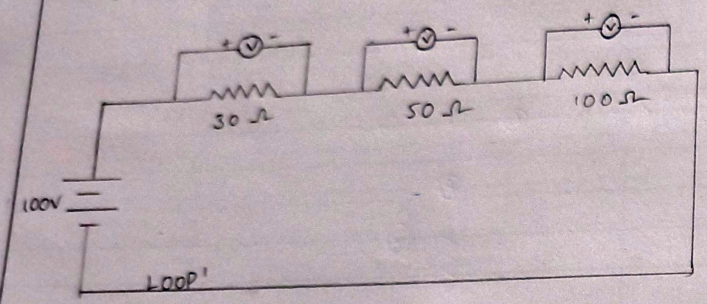




TABULATION:

KVL	Source (V)	$V_{30\Omega}$ (V)	$V_{50\Omega}$ (V)	$V_{100\Omega}$ (V)	$V_{30\Omega} + V_{50\Omega} + V_{100\Omega}$ (V)
Theoritical	100	16.68V	27.8V	55.6V	100.05
Theoritical	100	16.7V	27.8V	55.6V	100.01

Expt. No. 1

Page No. 1

VERIFICATION OF KIRCHOFF'S VOLTAGE LAW AND KIRCHOFF'S CURRENT LAW

AIM:

To verify Kirchoff's voltage Law for loop's and kirchoff's current law in node A for the following circuit.

APPARATUS REQURIED:

Laptop with Proteus software

THEORY:

According to kirchoff's voltage law: sum of potential raises are equal to sum of potential drop in a loop.

In LOOP 1 BY KVL:

$$V = V_1 + V_2 + V_3 \longrightarrow (1)$$

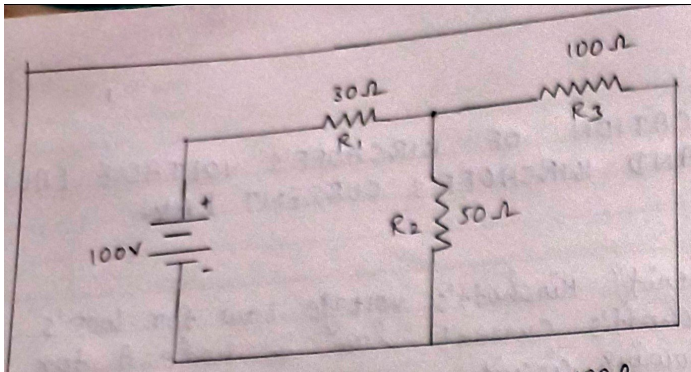
$$100 = IR_1 + IR_2 + IR_3$$

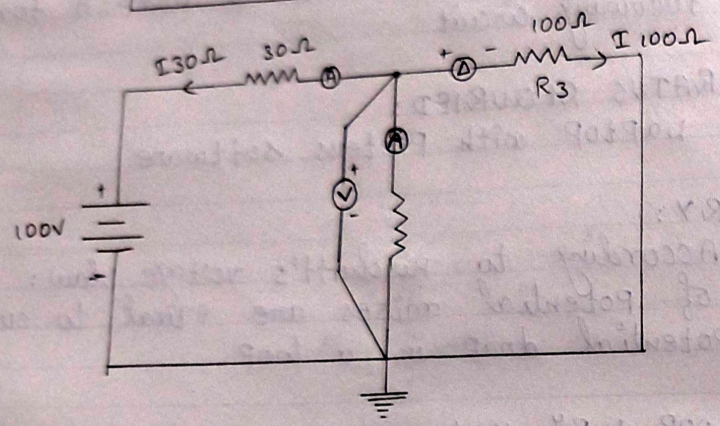
$$R_{eq} = R_1 + R_2 + R_3$$

$$R_{eq} = 30 + 50 + 100$$

$$R_{eq} = 180\,\Omega$$

$$I = \frac{V}{R_{eq}} = \frac{100}{180} = 0.556A$$





ABULATION :

KCL	I30Ω	I50Ω	I100Ω	VA
theoritical	-1.48A	1.85A	0.56A	52.63V
Theoritical	1.58A	1.05A	0.53A	52.6V

$V_1 = V_{30\Omega} = IR_1 = 0.556 \times 30 = 16.68V$
$V_2 = V_{50\Omega} = IR_2 = 0.556 \times 50 = 27.8V$
$V_3 = V_{100\Omega} = IR_3 = 0.556 \times 100 = 55.6V$

substitute V_1, V_2 and V_3 values in ①

$100 = 16.68 + 27.8 + 55.6$
$100V = 100V$
Hence proved.

KIRCHOFF'S CURRENT LAW:

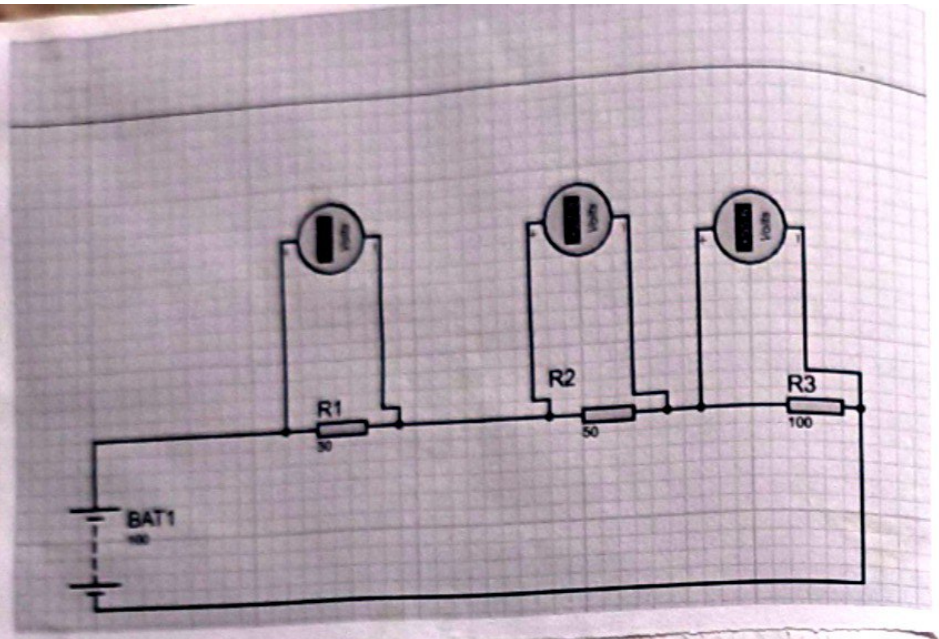
In a Junction, incoming current is equal to outgoing current.

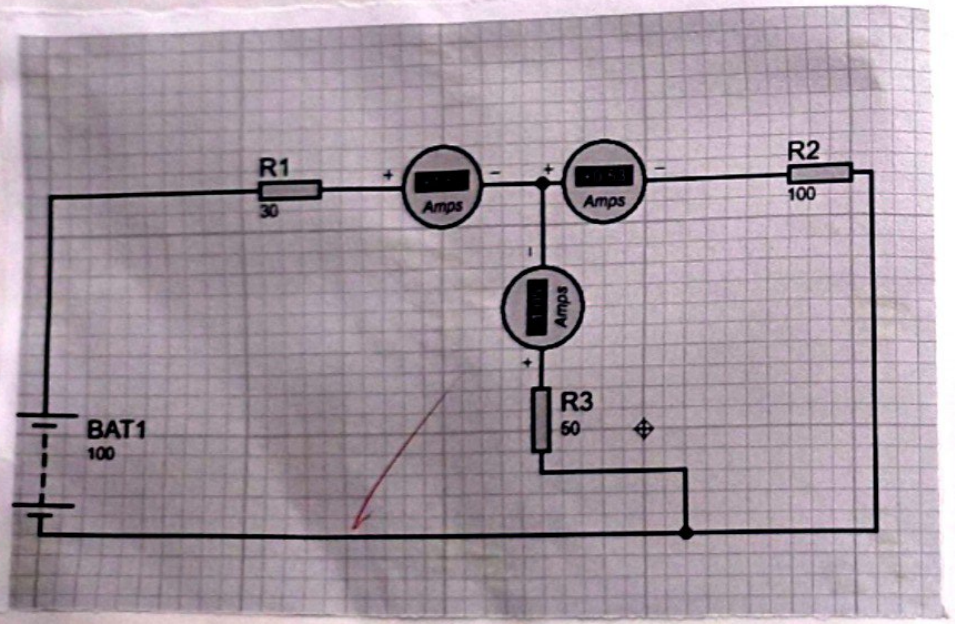
An node A :

$I_{30\Omega} + I_{50\Omega} + I_{100\Omega} = 0$
$V = 100/30 + V/50 + V/100 = 0$

$$V = \left[\frac{1}{30} + \frac{1}{50} + \frac{1}{100}\right] - \frac{100}{30} = 0$$

$V[0.0633] - 3.333 = 0$
$$V = \frac{3.333}{0.0633}$$





$V = 52.63\ V$

$I_{30\Omega} = \frac{V-100}{30} = \frac{52.63-100}{30} = -1.58A$

$I_{50\Omega} = \frac{V}{50} = \frac{52.63}{50} = 1.05A$

$I_{100\Omega} = \frac{V}{100} = \frac{52.63}{100} = 0.53A$

At node A :

$I_{30\Omega} + I_{50\Omega} + I_{100\Omega} = 0$

$-1.58 + 1.05 + 0.58 = 0$

Hence proved.

100

22/02/2026

Result :

Thus, the kirchoff's voltage law and kirchoff's current law has been verified successfully.